DLSS 2025 - Assignment 2 Convolutional Neural Network

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June 4, 2025

1 Introduction

The EuroSAT dataset consists of satellite images representing various land use and land cover types across Europe. It is introduced in the paper "EuroSAT: A Novel Dataset and Deep Learning Benchmark for Land Use and Land Cover Classification," in which the authors develop a state-of-the-art Convolutional Neural Network (CNN) to address the task of land use and land cover classification using satellite imagery. The authors report an accuracy of 98.57

The objective of this project is to design a CNN that achieves a solid accuracy of approximately 85% on this real-world dataset. This report outlines the process of data loading, describes the architecture used, and details the construction of the CNN. Furthermore, the CNN model is compared with a Multilayer Perceptron (MLP), and an analysis is provided on which categories were particularly difficult to classify and the possible reasons behind these challenges.

2 Results

2.1 Data

The dataset contains 27,000 satellite images, categorized into ten distinct land cover classes. These classes are: Highway, Forest, River, Permanent Crop, Residential, Herbaceous Vegetation, Industrial, Pasture, and Sea/Lake.

The first step involved loading the data, which was relatively straightforward. To gain an initial impression of the dataset, several randomly selected images were plotted along with their corresponding class labels (see Figure 1). For model development and evaluation, the dataset was split into training, validation, and test subsets. Specifically, 70% of the data was allocated for training, 15% for validation, and the remaining 15% for testing.

Unlike grayscale datasets such as Fashion-MNIST, the EuroSAT images are in color, which requires normalization across each of the RGB channels. In addition, data augmentation techniques were applied to the training set using a transformation pipeline to improve the model's generalization performance.

(a) Images of classes (b) Augmented images

Figure 1: Dataset samples and data augmentation examples

2.2 Models

The CNN model used in this project comprises a total of 1,114,858 trainable parameters. The architecture consists of a combination of different layer types. The model begins with a convolutional layer that accepts a three-channel input, corresponding to RGB images. If the dataset consisted of grayscale images, only a single

channel would be required. The first convolutional layer generates 32 feature maps using a 3×3 filter with padding to preserve the spatial dimensions of the input.

Following the convolution, a ReLU activation function is applied to introduce non-linearity—similar to the MLP models used previously. This structure is then repeated in a second convolutional layer. A 2×2 max pooling layer is incorporated to reduce the spatial dimensions by 50%, from 64×64 to 32×32. To improve generalization and reduce the risk of overfitting, a dropout layer with a dropout rate of 35% is added at the end of the first block.

A second convolutional block follows, mirroring the structure of the first block. The use of another max pooling layer further reduces the spatial dimensions to 16×16. Finally, the output is flattened and passed through a fully connected classification layer. This layer consists of ten output neurons, corresponding to the ten target classes, and performs the final class prediction.

The same model architecture is used in two separate training scenarios: one with the original (non-augmented) training data and one with augmented training data. The augmented model is expected to be more robust and achieve higher accuracy. Figure 2 illustrates an example of how a single image is altered through data augmentation, highlighting the changes introduced during this process.

Layer (type)	**Output Shape**	**Parameter**
Conv2d-1	[-1, 32, 64, 64]	896
ReLU-2	[-1, 32, 64, 64]	0
Conv2d-3	[-1, 32, 64, 64]	9,248
ReLU-4	[-1, 32, 64, 64]	0
MaxPool2d-5	[-1, 32, 32, 32]	0
Dropout-6	[-1, 32, 32, 32]	0
Conv2d-7	[-1, 64, 32, 32]	18,496
ReLU-8	[-1, 64, 32, 32]	0
Conv2d-9	[-1, 64, 32, 32]	36,928
ReLU-10	[-1, 64, 32, 32]	0
MaxPool2d-11	[-1, 64, 16, 16]	0
Dropout-12	[-1, 64, 16, 16]	0
Flatten-13	[-1, 16384]	0
Linear-14	[-1, 64]	1,048,640
ReLU-15	[-1, 64]	0
Dropout-16	[-1, 64]	0
Linear-17	[-1, 10]	650

Table 1: Neural Network Architecture Summary

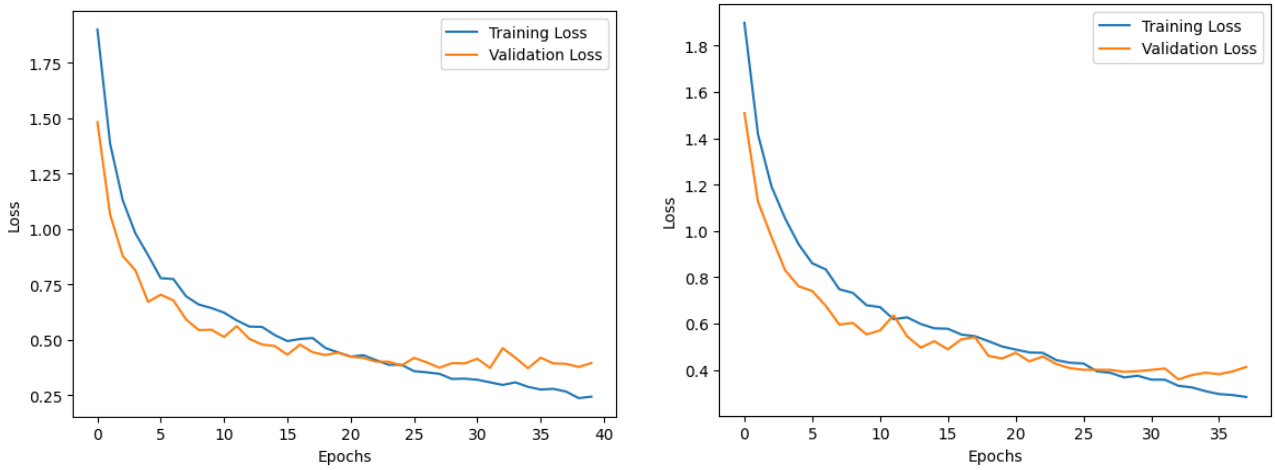
2.3 Training

During the training process of all models, early stopping is employed to prevent overfitting. The early stopping mechanism monitors the validation loss and saves the model whenever a decrease is observed. If the validation loss does not improve for five consecutive epochs—referred to as the patience—training is halted and the model is restored from the last saved checkpoint with the lowest validation loss. The hyperparameters are identical for both models. The learning rate is set to 0.001, and the L2 regularization (weight decay) is set to 0.0001. As this is a multiclass classification task, the loss function used is CrossEntropyLoss, which is suitable for such problems.

Figure 3 displays the learning curves for the CNN models trained with and without data augmentation. Both models exhibit initial signs of overfitting towards the end of the training process. Additionally, it is evident that early stopping was triggered in both cases, as neither model completed the full 50 training epochs. This indicates that the validation loss stopped improving before the maximum number of epochs was reached.

2.4 Evaluation

Table 2 presents the test set accuracy for both CNN models. The CNN trained without data augmentation achieves an accuracy of 89.61%. The table also includes class-wise performance, showing how well the model classifies images into the correct categories. In this model, classes 3, 6, and 8—corresponding to Highway, Permanent Crop, and River—have the lowest precision and are thus more challenging for the model to classify accurately.



(a) Training curves CNN (without augmentation)

(b) Training curves CNN (with augmentation)

Figure 2: Comparison of CNN training curves with and without data augmentation

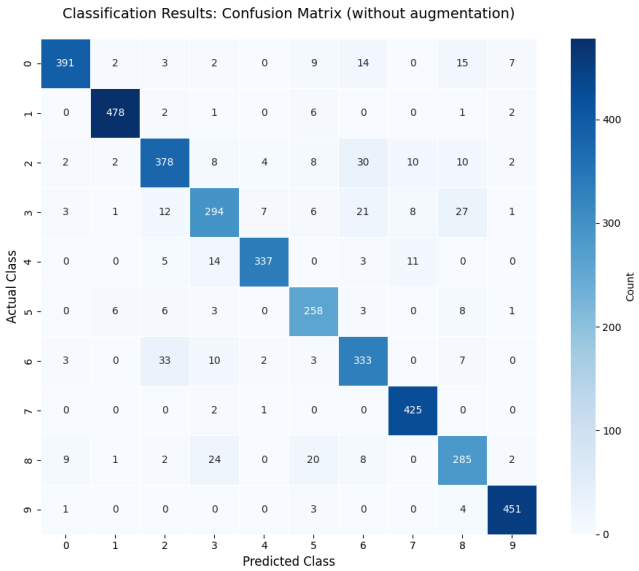


Figure 3: Heatmap: CNN without augmentation

The second model, trained with data augmentation, achieves a slightly lower accuracy of 88.97%. Here, the classes with the lowest precision are 3, 5, and 6 (Highway, Pasture, and Permanent Crop). In both models, the categories Highway and Permanent Crop consistently pose classification difficulties.

The lower overall accuracy of the augmented model is somewhat unexpected. Initially, it was hypothesized that data augmentation would lead to a more robust model with improved accuracy. This assumption does not appear to hold in this case. One possible explanation is that the augmentation applied was not strong or diverse enough to improve generalization. Alternatively, there may have been an issue in the data transformation process.

Figure 3 illustrates the confusion matrix for the model without augmentation, highlighting which classes are frequently misclassified. For instance, River images are often confused with Highway images and vice versa. Additionally, many images from class 2 (Herbaceous Vegetation) and class 6 (Permanent Crop) are misclassified, as well as class 3 (Highway) and class 6. These misclassifications are not entirely surprising, given that Highway and River images often share visual similarities, as do Herbaceous Vegetation and Permanent Crop. One possible explanation for the confusion between Highway and Permanent Crop is that highways often run through agricultural areas, producing visually similar satellite images.

Class	Precision	Recall	F1-Score
0	0.9560	0.8826	0.9178
1	0.9755	0.9755	0.9755
2	0.8571	0.8326	0.8447
3	0.8212	0.7737	0.7967
4	0.9601	0.9108	0.9348
5	0.8243	0.9053	0.8629
6	0.8083	0.8517	0.8294
7	0.9361	0.9930	0.9637
8	0.7983	0.8120	0.8051
9	0.9678	0.9826	0.9751
Accuracy			0.8961

(a) CNN without Data Augmentation

Class	Precision	Recall	F1-Score
0	0.9157	0.9074	0.9116
1	0.9599	0.9776	0.9687
2	0.8973	0.7313	0.8058
3	0.7898	0.7711	0.7803
4	0.9464	0.9541	0.9502
5	0.8400	0.8842	0.8615
6	0.7250	0.8900	0.7991
7	0.9507	0.9907	0.9703
8	0.8622	0.7664	0.8115
9	0.9805	0.9847	0.9826
Accuracy			0.8897

(b) CNN with Data Augmentation

Table 2: Comparison of CNN Classification Reports

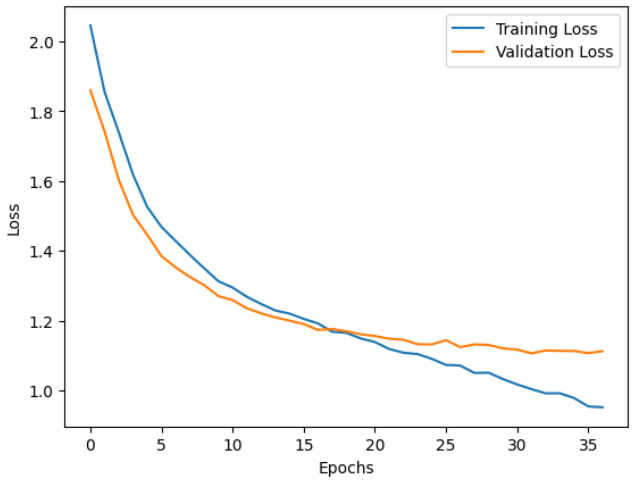


Figure 4: MLP training curve

3 Conclusions and Discussion

In this project, I implemented the same Convolutional Neural Network (CNN) architecture twice—once with and once without data augmentation. Initially, I hypothesized that the model trained with augmented data would outperform the one trained without. However, this was not the case. This unexpected result suggests that the augmentation pipeline should be reviewed for possible issues or adjusted to introduce more effective variations.

I presented the implementation of the models, evaluated their performance, and analyzed which classes were most difficult to classify correctly. In both models, certain classes were frequently confused with others, particularly Highway, Permanent Crop, and River, which often share visual similarities in satellite imagery.

4 Bonus 1

For this task, I also trained a Multilayer Perceptron (MLP) on the EuroSAT dataset. The MLP uses a simple architecture with two hidden layers and a total of 6,457,482 trainable parameters. As observed with the CNNs, the MLP also shows signs of overfitting—perhaps even more pronounced. The MLP achieves an accuracy of 60.28%, which is significantly lower than that of the CNNs.

This outcome is not surprising, as CNNs are specifically designed for image classification tasks and are generally more effective at capturing spatial hierarchies in visual data. Nevertheless, the results also demonstrate that an MLP can handle image data to some extent, even though it is not optimized for this type of task.

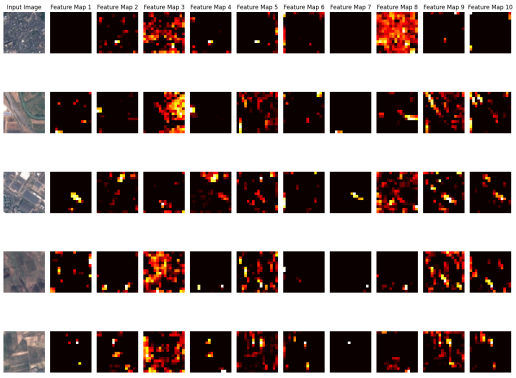


Figure 5: Feature Map

5 Bonus 2

In this section, we examine the feature maps utilized by the CNN. It becomes evident that the feature maps are not equally activated across all input images. For instance, feature map eight is strongly activated for the first and third images. The first image belongs to the Residential class, while the third represents an Industrial area. Given that these two images share more visual similarities than the other examples in the figure, the similar activation of this feature map suggests that it is detecting shared features between them.

When the same feature map is activated across different images, it indicates that those images exhibit similar patterns or structures in the areas where the activation occurs. Additionally, we observe that some feature maps are not activated at all. This does not necessarily mean that these feature maps are inactive or "dead" in general, but rather that they are not triggered by the particular set of images shown here. It is likely that elsewhere in the dataset, there are images that activate these maps, indicating their relevance for capturing other specific features.